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Case study

Using jump lists to identify fraudulent documents

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ABSTRACT

Jump lists show the file opening activity of a computer user. When a computer user wants to know the most recent file they opened, a jump list can provide that information. Windows 7 displays jump lists for recently used files, but more importantly for investigators, it also records hidden jump list artifacts. These hidden jump list artifacts reveal the complete trail a fraudster follows in creating fraudulent documents or to perform other illegal activities when using their computers. Such jump list artifacts can remain on the computer's drives for years. The paper describes a method that can be used to identify artifacts and their potential for use as forensic evidence in a financial fraud case.

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Today business data is stored in an electronic format. For that reason, a fraud investigation cannot be successfully prosecuted without thoroughly investigating electronic data. In many legal cases, the smoking gun in the case is electronic evidence that is forensically collected by investigators. Such electronic evidence may include e-mail messages, metadata in the computer, or web browser histories. Electronic evidence contributes to a case in a way that paper documents or witness testimony cannot because a computer methodically records all digital activities. The collection of electronic evidence involves its acquisition, investigation, analysis, and event reporting. Those services from a computer forensic analyst may cost about $350 per hour. Depending on the nature of the case, total costs can quickly reach $10,000 or more.

Therefore, if there is a reliable forensic method available with the potential for decreasing investigative costs, it should be a consideration in a digital investigation. JumpLister is a free software program that has the potential for reducing such investigative expenses.[1] The paper reviews the forensic value of jump lists and JumpLister using a hypothetical financial fraud case.[2]

Simple jump lists are used to identify the files that have been most recently opened by a computer user. For example, the Windows 7 operating system uses a jump list to help the user see the most recent files they have opened (Jump Lists, Accessed August 27, 2012). By clicking on the Desktop's Start button and then placing the cursor over the Microsoft Word 2010 icon, a list of recently opened programs is shown. Thus, a "jump list" is a record of the user's recent activity on a PC. The Windows 7 jump list eventually changes as files are dropped off the list and new files are added.

Beyond these jump lists, the Windows 7 system also records all activities in a set of hidden jump list artifacts

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[1] The author has no relationship, business or otherwise, with Mark Woan, the creator of the JumpLister program described in this paper.

[2] For an overview of jump list analysis see the Windows Incident Response Blog (Carvey, 2011) at: http://windowsir.blogspot.com/2011/12/jump-list-analysis.html.

within the user's PC.[3] The artifacts track all user activity on the PC, and they show all the supporting activity that was used create a document such the use of a jpeg or template. These jump lists are artifacts in the operating system and their recovery can be useful in a forensic investigation. Barnett (no date) worked with images to identify Windows 7 jump list characteristics. He noted that very little has been written about the value of jump lists in forensic investigations. The characteristics of jump list artifacts in Notepad and MS Paint were analyzed by Lyness (2012). For an example of the raw data analysis of jump list artifacts, see Larsen (no date) who applies utilities such as OffVis (http://blogs.technet.com/b/srd/archive/2009/09/14/offvis-updated-office-file-format-training-video-created.aspx) and MiTec's Structured Storage Viewer (http://www.mitec.cz/ssv.html) to carve and parse jump list data.

It should be noted that unlike recent files jump lists obtainable from using the Windows 7 Start button, the data about these artifacts is permanent. In fact, these data artifacts are not automatically deleted even if the user deletes the files and the software used to create the file.[4] The artifacts show connections to removable devices that have been connected to the PC such as flash drives. The information in the artifact identifies the drive location in the PC where the file was created (Birth Volume) and the new location if the file has been copied or moved (New Volume). Created, accessed, and modified dates are part of the information provided about the file. These dates are unlike those dates shown when the "Properties" of a file are shown as the dates in the artifacts cannot be modified by the user. Thus the data provides an investigator with a timeline of software and files usage.

1. Creation of a fraudulent document and its forensic investigation

In Fig. 1, a fraudulent purchase order has been created by Larry Smith, Controller.[5] Larry had authority to clear the purchase order through payables and ensure it would be paid. Larry wanted to have the laptops he ordered sent to his brother rather than to his employer. For our purposes, the purchase order was purposely created in a fashion whereby it would be quickly identified as suspicious with data mining software such as IDEA or ACL when authorized address lists are checked.[6] It can be seen in Fig. 1 that Mark Smith's address is a red flag indicating the purchase order is probably fraudulent. But what if the suspect, Larry Smith, denies that he fabricated the document? If additional forensic data can be collected, it will make his denial less plausible.

In order to fabricate the purchase order, Larry downloaded a logo from Southeastern Oklahoma University's (SE) website.[7] To create any official looking document, a first step is to place the correct company logo at the top of the document. Such a logo can come from the Internet or the company's internal database. In this case, Larry downloaded the image from the Internet as he believed his activities would be less likely to be detected. He knew his activities in the purchase order database were closely logged and a review of those logs might raise the suspicions of the financial auditors.

After downloading the logo, Larry placed the image in the Pictures folder on his work PC. Next, he downloaded a purchase order template from those available in MSWord 2010. He then completed the fabricated purchase order so that the laptops would be delivered to his brother Mark Smith, and moved the fraudulent purchase order into an unrelated folder where he hoped it would be hard to find. He then sent the purchase order to the vendor. As the university had administrative facilities in Ada, Oklahoma, the vendor would not suspect that the invoice was fraudulent. Afterward, Larry deleted the purchase order from his computer. These were the steps Larry followed as he fabricated the fraudulent purchase order.

Financial auditors use data mining software to review thousands of invoices, purchase orders and other documents in the accounting databases. Anomalies in these documents will red flag suspicious documents. The addresses on the purchase order raised a red flag for the auditor and Larry was asked about the purchase order. It should be noted that the use of IDEA and ACL to open a file will corrupt the last accessed and modified dates in the file's properties as viewed in MSWord, for example. Thus, Larry can insist that he was not using the PC at those times and someone else created the fraudulent purchase order. Larry immediately denied having any knowledge about the purchase order, i.e., "someone else did it."

[3] Each of the artifacts in the hidden jump list begins with a long string of characters called the application ID (AppID) used to identify the specific software application opened to access and create a file. For example, 1b4dd67f29cb1962.automaticDestinations-ms = **Windows Explorer** Pinned and Recent and adecfb853d77462a.automaticDestinations-ms = **Word 2007** Pinned and Recent. Pinned means it is on the taskbar. It is important to know the software used to create a file as recovery information about a specific file is tied to the software and its corresponding AppID. When artifacts are disclosed, they only show the AppID. For that reason, the investigator needs to correlate an AppID with their software application. A list of AppIDs and their related software application is available at ForensicsWiki, as well as one on the ForensicArtifacts site. Not all AppIDs and their software applications are listed, but any AppID string can be searched using a search engine such as Google.

[4] Manually deleting an artifact will receive evidence of its deletion (Lyness, 2012).

[5] In our example, the AppID is *a7bd71699cd38d1c* refers to a file created under Microsoft Word 2010. For additional information regarding the creation of fraudulent financial documents see Journey Into Incidence Response Blog (Harrell, 2012) at http://journeyintoir.blogspot.com/2012/03/microsoft-word-jump-list-tidbit.html.

[6] IDEA and ACL software packages are used by accounting auditors in detecting fraudulent activities by identifying unusual activities in the review of thousands of accounting records. The packages are also used for reconciliations, and evaluating operational performance. For information about these packages see: http://www.audimation.com/idea.html and http://www.acl.com/.

[7] This example is purely hypothetical, and it is not based on a real-life case. Any logo could have been used to illustrate the creation of a fraudulent purchase order.

SOUTHEASTERN OKLAHOMA STATE UNIVERSITY

PURCHASE ORDER

459 Oak Street
Ada, Oklahoma
5806725555

TO:
Larry Smith
Physics Dept.
5806725555

SHIP TO:
Mark Smith
459 Oak Street
Ada, Ok 75234
5806725555

P.O. NUMBER:
12-3756

P.O DATE	REQUISITIONER	SHIPPED VIA	F.O.B. POINT	TERMS
5/26/2011	Larry Smith	UPS	Ada, OK	none

QTY	UNIT	DESCRIPTION	UNIT	TOTAL
3	H768s	HP Computer Model	$615	$1845

SUBTOTAL	1845
SALES TAX	0
SHIPPING AND	75
OTHER	
TOTAL	**$1920**

1. Please send two copies of your invoice.
2. Enter this order in accordance with the prices, terms, delivery method, and specifications liste above.
3. Please notify us immediately if you are unable ship as specified.
4. Send all correspondence to:
   Mark Smith
   459Oak Street
   Ada, Oklahoma
   5806725555

Authorized byLarry Smith 5/26/2012

Fig. 1. The fraudulent purchase order created by Larry Smith.

2. The Jumplister

In order to further identify Larry's activities on his work computer, it is necessary to find the hidden artifacts in his Windows 7 jump list. The dates on the original file within the artifacts do not change and thus provide solid forensic evidence of the actual date the file was created and when it was last accessed. In addition, the artifacts show the exact trail that Larry used to create the fraudulent purchase order. The format of these jump list files need to be cleaned up or "parsed" as they are not recorded in a readable format. An excellent free software utility program that can be used to parse the data is JumpLister by Mark Woan (http://www.woanware.co.uk/?page_id=266). The JumpLister program is easy to use, and it quickly identifies the forensic information needed to

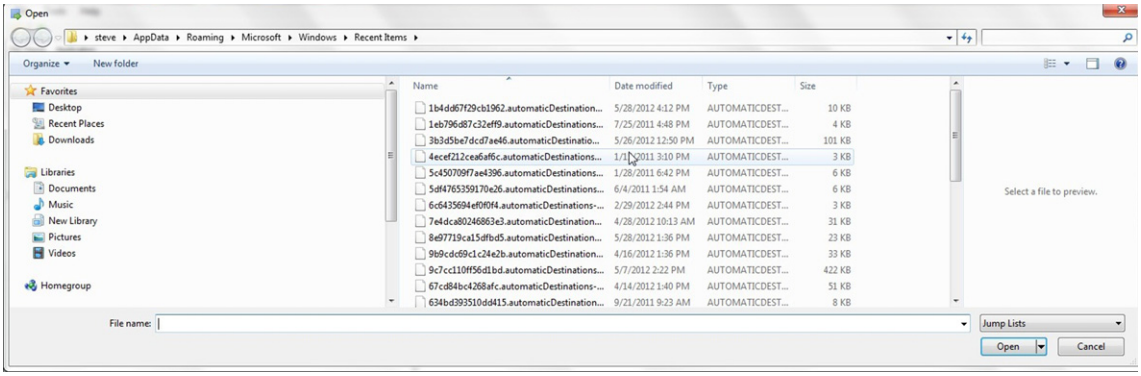


Fig. 2. The opening screen on JumpLister shows the list of artifacts and the associated date. Within each artifact, there is a list of secondary files recording the recent activity that occurred on various dates.

show how the fraudulent purchase order was created as well as the unchanged dates and time these activities were performed. JumpLister's opening screen is shown in Fig. 2. JumpLister stores the artifacts in Automatic Destination by the software program used to create the file. Each Automatic Destination string is associated with a specific software program. In Fig. 2, it can be seen that there is a list of dates for each file. The objective is to find the dates that are the same as when the purchase order was created, i.e., May 26, and then open those artifacts to find information about how the purchase order was produced.

To identify the steps followed in creating the purchase order, the artifacts for May 26 are searched for activities related to the creation of the fraudulent purchase order. Several files are found. When working with large records, records within the artifacts can be exported to a Word file and searched using a keyword search based on the date of interest to the investigator. Here it is was not necessary to export the records of interest. Once a May 26 artifact is located, it is opened and the JumpLister program identifies a list of numbered records and their dates. Without the JumpLister program this information is not easily available. At this point, it is necessary to manually search each of the numbered records to find the one related to the fraudulent purchase order.

The first one that is found is related to Larry's downloading of the SE logo as seen in Fig. 3. In Fig. 3, it can be seen that each of the records is numbered. The activity related to the SE logo is number 106. The activities related to the logo show that it was "hidden" in an unrelated folder

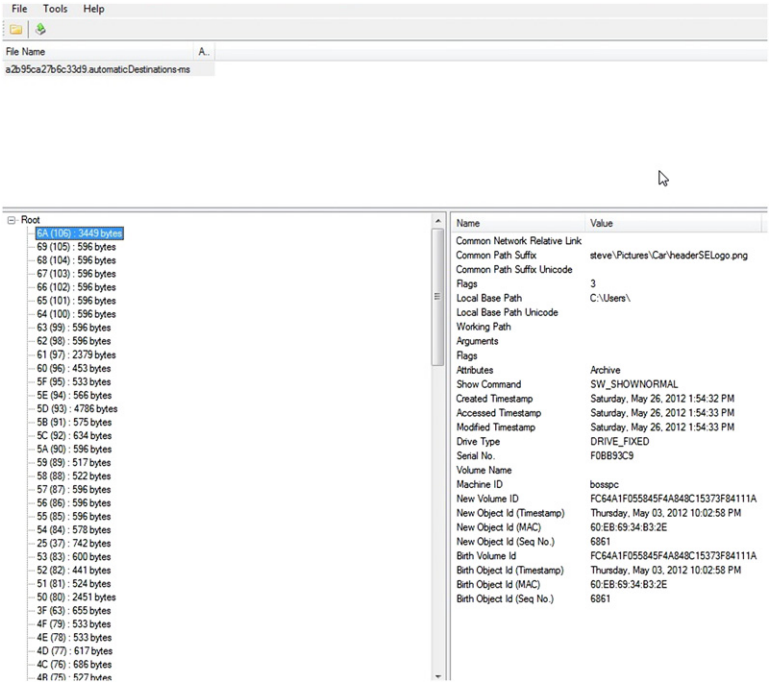


Fig. 3. Jump list showing downloaded SE logo (#106) was "hidden" in a Picture folder titled "Car".

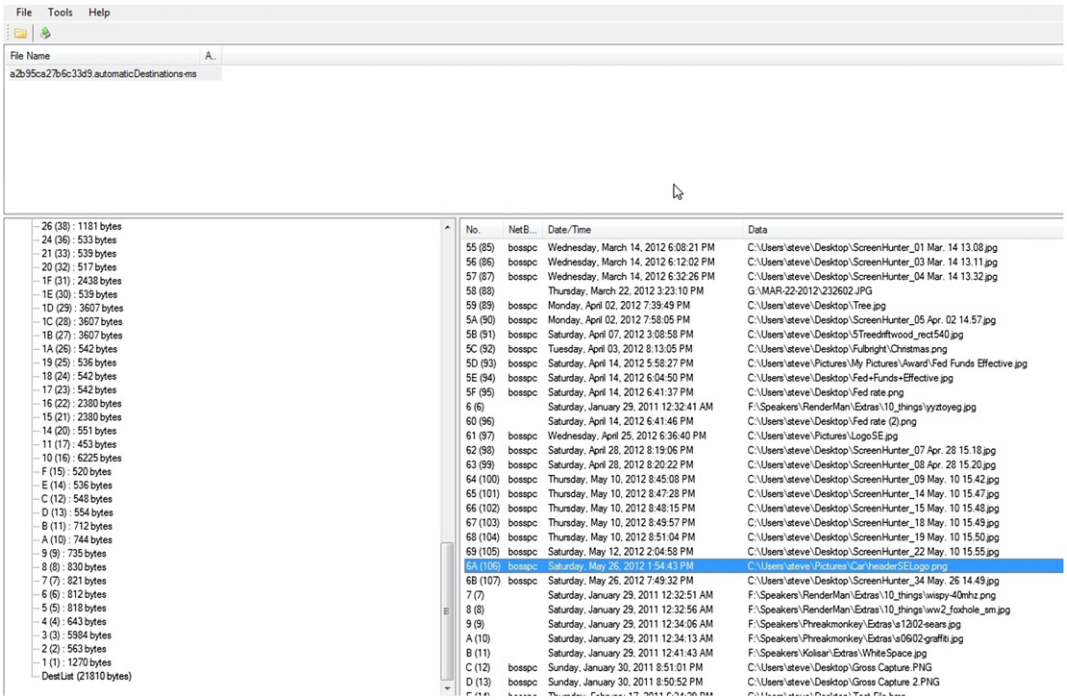


Fig. 4. Using the destination list to find activity on Larry Smith's computer relating to the falsified purchase order created on May 26, 2012.

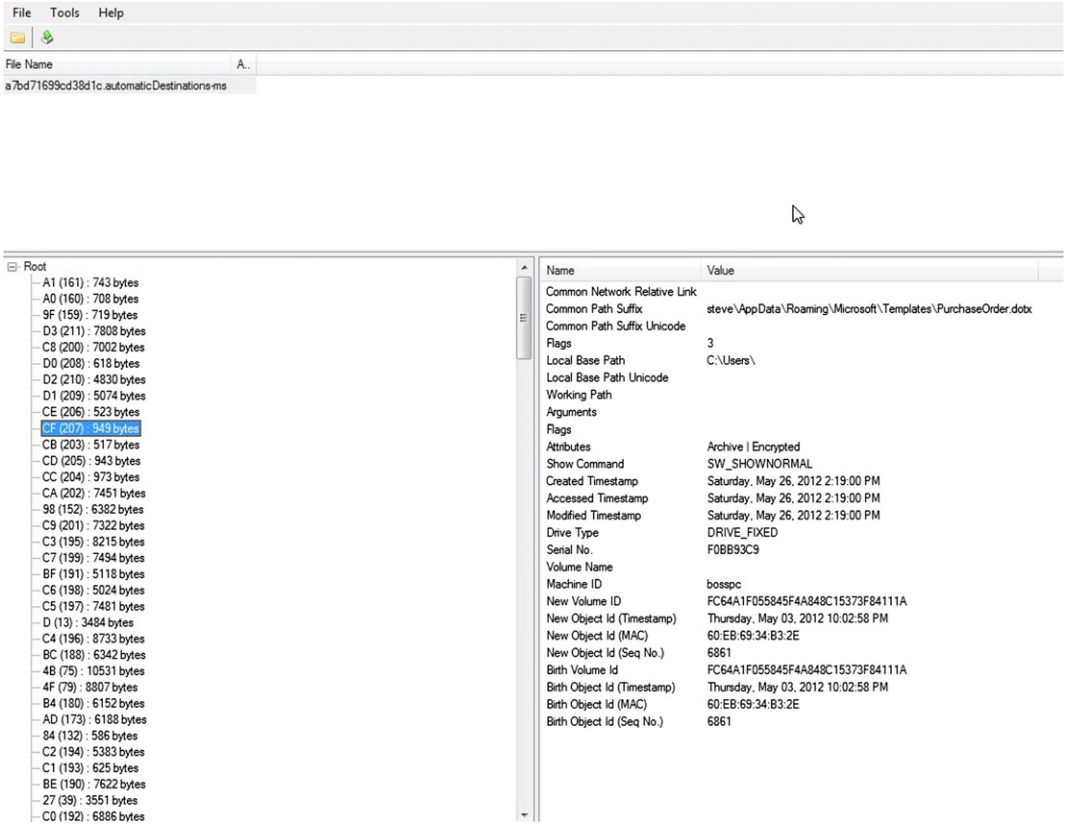


Fig. 5. The artifacts showing Larry Smith downloaded a Microsoft template to create a fraudulent purchase order.

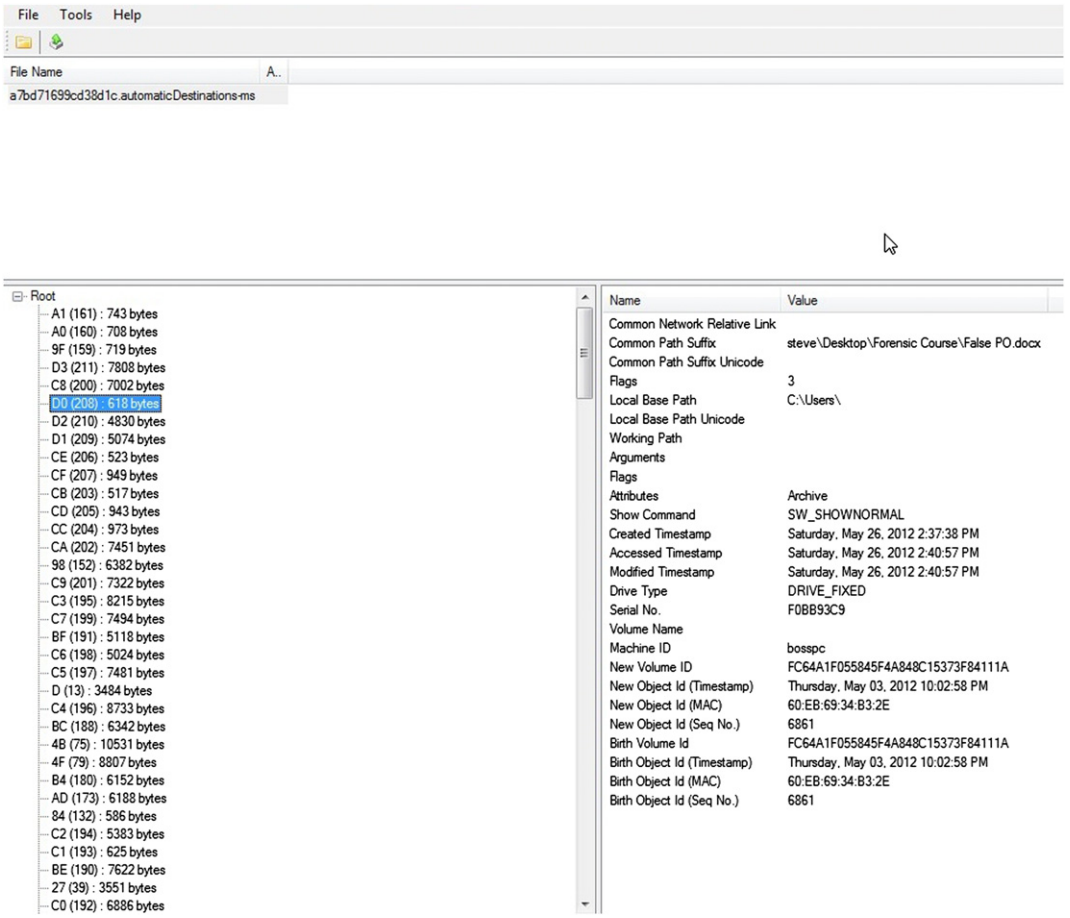


Fig. 6. After the false purchase order is completed it is placed in a folder called forensic course on the desktop.

on the desktop for later use.[8] The dates when this activity occurred are shown in Fig. 3.[9]

If the identification of the artifacts becomes more complicated, the Destination List (DestList) can be used to locate the correct file. By selecting one of the "automatic destination" artifacts as shown in Fig. 2, a root list will appear and at the bottom of the root list is a DestList as shown on the left of the screen in Fig. 4. In Fig. 4, it can be seen that the DestList is selected and on the right of the screen a file dated May 26 is highlighted. This is the file of interest to the investigator. Once a DestList is selected, it separately shows all the files in each artifact. The long list of files on the right of the screen shows the dates they were created and allows the investigator to identify a date of interest to the investigation. In the example, this is how the investigator was able to identify artifact #106. These files showing user activity remain present for years, and opening them does not corrupt the actual files or change any metadata.

At this point, an investigator would search the artifacts for other activities occurring on May 26. The next piece of the puzzle is shown in Fig. 5. It can be seen that Larry Smith downloaded a purchase order template from Microsoft. The specific artifact recorded in the computer's jump list is #207. On the right side of the screen, in Fig. 5, is the information recorded by jump list #207.[10] Clearly the path, Common Path Suffix, shows that a template was downloaded ("roaming") and later used by Larry Smith to create the fraudulent purchase order. The template could have been taken from the university's database, but Larry believed the electronic trail would be better hidden if he used a template from an outside source. With the use of JumpLister, these activities are clearly evident and help to show that Larry did create the purchase order. JumpLister shows the location where the purchase order was stored in the computer. The dates help to show that Larry was not on vacation at these times even if the dates under a file's MS Word's *Properties* have been falsified to show dates when Larry was not at work, for example.

[8] Steve\Pictures\Car\headerSELogo/png shows a folder (headerSELogo/png) was placed in a folder called "Car" in the Pictures Folder.

[9] In Fig. 3, the machine's ID is "bosspc" which identifies the computer used to create the purchase order.

[10] Just as artifact #106 was identified by its May 26 date, the new artifact #102 is also identified by its date on a Destination List. The Destination List for #102 is not shown.

The last screen shot from JumpLister, Fig. 6, shows that the fraudulent purchase order was completed and stored in a folder, called Forensic Course, on the Desktop. Larry has stored the completed purchase order in an unrelated folder to prevent it from being found. Even if the purchase order file is later deleted, JumpLister would still show all the activities related to the creation of the fraudulent purchase order. After Larry sent the purchase order to the vendor, he immediately deleted the file from his PC.

Larry's activities could have been identified if an image of his computer's hard drive had been made and then analyzed for the files that Larry tried to hide. The results would identify the same locations within the computer as identified by Jumplister. Prior to hiring a forensic computer analyst and using a program like EnCase and beginning an intensive investigation, it is possible to collect forensic evidence that identify Larry's activities and make it more difficult for him to say "someone else did it."

Although this example dealt with fraudulent financial information, jump lists can be used in many other situations where it is important to determine PC usage patterns. Beyond providing information as to whether a user opened a specific application or uploaded or downloaded images, a jump list can identify whether a flash drive or DVD was put into the system. As shown here, the data in a jump list provides information as to whether a user purposefully tried to hide incriminating evidence by deleting files. But AppIDs are available for browsers, media players, IM communication, file sharing, FTP and IRC, newsreaders and system cleaners such as CCleaner. Thus, the forensic investigator is able to trace the activities that occurred on a PC in a much wider range than those encountered in creating a fraudulent financial document.

3. Summary

The case presented here was a financial fraud, but jump list artifacts recorded within Windows 7 extend beyond the creation of fraudulent purchase invoices. In each of those instances, the initial use of JumpLister can increase the likelihood of quickly identifying the perpetrators of fraudulent or destructive acts. JumpLister is one method to identify forensic information without a great deal of difficulty and without forensically damaging electronic evidence. Such digital evidence remains accessible years after an event has occurred. Programs such as JumpLister make it much more difficult for a suspect to deny their illegal actions once they are presented with a report identifying each step of their activities.

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